

How Long Until It Works

- Onset of therapeutic actions in depression with transdermal administration is usually not immediate, but often delayed 2–4 weeks or longer following adequate dosing
- If it is not working for depression within 6–8 weeks, it may require a dosage increase or it may not work at all
- May continue to work in depression for many years to prevent relapse of symptoms
- Can enhance the actions of levodopa in Parkinson's disease within a few weeks of initiating oral dosing
- Theoretical slowing of functional loss in both Parkinson's disease and Alzheimer disease is a provocative possibility under investigation and would take many months or more than a year to observe

If It Works

- The goal of treatment in depression is complete remission of current symptoms as well as prevention of future relapses
- Treatment of depression most often reduces or even eliminates symptoms, but not a cure since symptoms can recur after medicine stopped
- Continue treatment of depression until all symptoms of depression are gone (remission)
- Once symptoms of depression are gone, continue treating for 1 year for the first episode of depression

- For second and subsequent episodes of depression, treatment may need to be indefinite
- Continue use in Parkinson's disease as long as there is evidence that selegiline is favorably enhancing the actions of levodopa
- Use of selegiline to slow functional loss in Parkinson's disease or Alzheimer disease would be long-term if proven effective for this use

If It Doesn't Work

- Many depressed patients have only a partial response where some symptoms are improved but others persist (especially insomnia, fatigue, and problems concentrating)
- Other depressed patients may be nonresponders, sometimes called treatment-resistant or treatment-refractory
- Some depressed patients who have an initial response may relapse even though they continue treatment, sometimes called "poop-out"
- For depression, consider increasing dose, switching to another agent, or adding an appropriate augmenting agent, psychotherapy, and evaluation for another diagnosis or for a comorbid condition (e.g., medical illness, substance abuse, etc.)
- Some patients may experience apparent lack of consistent efficacy due to activation of latent or underlying bipolar disorder, and require antidepressant discontinuation and a switch to a mood stabilizer
- Use alternate treatments for Parkinson's disease or Alzheimer disease



Best Augmenting Combos for Partial Response or Treatment Resistance

* Augmentation of selegiline has not been systematically studied in depression, and this is something for the expert, to be done with caution and with careful monitoring

- A stimulant such as d-amphetamine or methylphenidate (with caution and by experts only as use of stimulants with selegiline is listed as a warning; may activate bipolar disorder and suicidal ideation; may elevate blood pressure)
- Lithium
- Mood-stabilizing anticonvulsants
- Atypical antipsychotics (with special caution for those agents with monoamine reuptake blocking properties, such as lurasidone, ziprasidone, and zotepine)
- Carbidopa-levodopa (for Parkinson's disease)

Tests

- Patients should be monitored for changes in blood pressure
- Although preliminary evidence from clinical trials suggests little or no weight gain, nonselective MAOIs are frequently associated with weight gain

Side Effects

How Drug Causes Side Effects

- At recommended transdermal doses, norepinephrine, serotonin, and dopamine increase in parts of the brain and at receptors other than those that cause therapeutic actions
- At high transdermal doses, loss of tissue selectivity and loss of MAO-A sparing actions in the gut may enhance the possibility of dietary tyramine interactions if MAO-B inhibition occurs in the gut
- At recommended oral doses, dopamine increases in parts of the brain and body and at receptors other than those that cause therapeutic actions
- Side effects are generally immediate, but immediate side effects often disappear in time

Notable Side Effects

- Transdermal: application site reactions, headache, insomnia, diarrhea, dry mouth
- Oral: exacerbation of levodopa side effects, especially nausea, dizziness, abdominal pain, dry mouth, headache, dyskinesia, confusion, hallucinations, vivid dreams



Life-Threatening or Dangerous Side Effects

- Transdermal: hypertensive crisis was not observed with preliminary experience in clinical trials, even in patients who were not following a low-tyramine diet
- Oral: hypertensive crisis (especially when MAOIs are used with certain tyramine-containing foods or prohibited drugs) –reduced risk

at low oral doses compared to nonselective MAOIs

- Theoretically, when used at high doses may induce seizures and mania as do nonselective MAOIs

Weight Gain



- Transdermal: reported but not expected; some patients may experience weight loss
- Oral: occurs in significant minority

Sedation



- Reported but not expected
- Can be activating in some patients

What to Do About Side Effects

- Wait
- Wait
- Wait
- Lower the dose
- Switch after appropriate washout to an SSRI or newer antidepressant (depression)
- Switch to other antiparkinsonian therapies (Parkinson's disease)

Best Augmenting Agents for Side Effects

- Trazodone (with caution) for insomnia in depression
- Benzodiazepines for insomnia in depression

Dosing and Use

Usual Dosage Range

- Depression (transdermal): 6 mg/24 hours–12 mg/24 hours
- Depression (oral): 30–60 mg/day
- Parkinson's disease/Alzheimer disease: 5–10 mg/day

Dosage Forms

- Transdermal patch 20 mg/20 cm² (6 mg/24 hours), 30 mg/30 cm² (9 mg/24 hours), 40 mg/40cm² (12 mg/24 hours)
- Capsule 5 mg
- Tablet 5 mg scored
- Orally disintegrating tablet 1.25 mg

How to Dose

- Depression (transdermal): initial 6 mg/24 hours; can increase by 3 mg/24 hours every 2 weeks; maximum dose generally 12 mg/24 hours
- Parkinson's disease: initial 2.5 mg/day twice daily; increase to 5 mg twice daily; reduce dose of levodopa after 2–3 days

Dosing Tips

- Transdermal patch contains 1 mg of selegiline per 1 cm² and delivers approximately 0.3 mg of selegiline per cm² over 24 hours
- Patch is available in three sizes –20 mg/20 cm² , 30 mg/30 cm² , and 40 mg/40 cm² –that deliver doses of approximately 6 mg, 9 mg, and 12 mg, respectively, over 24 hours
- At 6 mg/24 hours (transdermal) dietary adjustments are not generally required
- Dietary modifications to restrict tyramine intake from foods are recommended for doses above 6 mg/24 hours (transdermal)
- Transdermal patch should only be applied to dry, intact skin on the upper torso, upper thigh, or outer surface of the upper arm
- New application site should be selected for each day; patch should be applied at approximately the same time every day; only one patch should be applied at a time; patches should not be cut
- In cases of topical allergic reaction, it is likely due to adhesive, so consider cutting off adhesive and use nonallergic tape to hold patch on
- Avoid touching the exposed (sticky) side of the patch, and after application, wash hands with soap and water; do not touch eyes until after hands have been washed
- Heat could theoretically increase the amount of selegiline absorbed from the transdermal patch, so patients should avoid exposing the application site to external sources of direct heat (e.g., heating pads, prolonged direct sunlight)

- Although there is theoretically a 3-day reservoir of drug in each patch, multiday administration from a single patch is generally not recommended and has not been tested; because of residual drug in the patch after 24 hours of administration, discard used patches in a manner that prevents accidental application or ingestion by children, pets, or others
- For Parkinson's disease, oral dosage above 10 mg/day generally not recommended
- Dosage of carbidopa-levodopa can at times be reduced by 10–30% after 2–3 days of administering oral selegiline 5–10 mg/day in Parkinson's disease
- At doses above 10 mg/day (oral), selegiline may become nonselective and inhibit both MAO-A and MAO-B
- At doses above 30 mg/day (oral), selegiline may have antidepressant properties
- Patients receiving high oral doses may need to be evaluated periodically for effects on the liver
- Doses above 10 mg/day (oral) may increase the risk of hypertensive crisis, tyramine interactions, and drug interactions similar to those of phenelzine and tranylecypromine

Overdose

- Overdose with the transdermal formulation is likely to produce substantial amounts of MAO-A inhibition as well as MAO-B

inhibition, and should be treated the same as overdose with a nonselective oral MAOI

- Dizziness, anxiety, ataxia, insomnia, sedation, irritability, headache; cardiovascular effects, confusion, respiratory depression, coma

Long-Term Use

- Long-term use has not been systematically studied although generally recommended for chronic use as for other antidepressants

Habit Forming

- Lack of evidence for abuse potential with transdermal selegiline despite its metabolism to l-amphetamine and l-methamphetamine
- Dependence to other MAOIs reported but rare

How to Stop

- Transdermal: MAOI slowly recovers over 2–3 weeks after patch removed
- Oral: generally no need to taper, as the drug wears off slowly over 2–3 weeks

Pharmacokinetics

- Clinical duration of action may be up to 14 days due to irreversible enzyme inhibition
- Major metabolite of selegiline is desmethylselegiline; other metabolites are l-methamphetamine and l-amphetamine

- Because first-pass metabolism is not extensive with transdermal dosing, this results in notably higher exposure to selegiline and lower exposure to metabolites as compared to oral dosing
- With transdermal selegiline, 25–30% of selegiline content is delivered systemically over 24 hours from each patch
- Mean half-life of transdermal selegiline is approximately 18–25 hours
- Steady-state mean elimination half-life of oral selegiline is approximately 10 hours



Drug Interactions

- Many misunderstandings about what drugs can be combined with MAOIs
- Theoretically and especially at high doses, selegiline could cause a fatal “serotonin syndrome” when combined with drugs that block serotonin reuptake, so do not use with a serotonin reuptake inhibitor for up to 5 half-lives after stopping the serotonin reuptake inhibitor (i.e., “wash-in” of selegiline should be about 1 week after discontinuing most agents [except 5 weeks or more after discontinuing fluoxetine because of its long half-life and that of its active metabolite]) (see [Table 1](#) after Pearls)
- When discontinuing selegiline (“wash-out” period), wait 2 weeks before starting another antidepressant in order to allow enough time for the body to regenerate MAO enzyme
- Transdermal: no pharmacokinetic drug interactions present in studies with alprazolam, ibuprofen, levothyroxine, olanzapine, risperidone,

and warfarin

- Tramadol (a serotonin reuptake inhibitor) may increase the risk of seizures in patients taking an MAOI
- Hypertensive crisis with headache, intracranial bleeding, and death may result from combining nonselective MAOIs with sympathomimetic drugs (e.g., amphetamines, methylphenidate, cocaine, dopamine, epinephrine, norepinephrine)
- Do not combine with another MAOI, alcohol, or guanethidine
- Adverse drug reactions can result from combining MAOIs with tricyclic/tetracyclic antidepressants and related compounds, including carbamazepine, cyclobenzaprine, and mirtazapine, and should be avoided except by experts
- Carbamazepine increases plasma levels of selegiline
- MAOIs in combination with spinal anesthesia may cause combined hypotensive effects
- Combination of MAOIs and CNS depressants may enhance sedation and hypotension

Table 1. Drugs contraindicated due to risk of serotonin syndrome/toxicity

Do Not Use:			
Antidepressants	Drugs of Abuse	Opioids	Other
SSRIs	MDMA (ecstasy)	Meperidine	Non-subcutaneous sumatriptan
SNRIs	Cocaine	Tramadol	Chlorpheniramine

Do Not Use:

Antidepressants	Drugs of Abuse	Opioids	Other
Clomipramine	Methamphetamine	Methadone	Brompheniramine
Imipramine	High-dose or injected amphetamine		Dextromethorphan
St. John's wort			Lumateperone
			Ziprasidone

 Other Warnings/Precautions

- Ingestion of a “high-tyramine meal” is generally defined as 40 mg or more of tyramine in the fasted state
- Studies show that 200–400 mg of tyramine in the fasted state (and even more ingestion of tyramine in the fed state) may be required for a hypertensive response with administration of the low-dose transdermal patch (6 mg/24 hours); thus, no dietary precautions are required at this dose
- Tyramine sensitivity of the low-dose transdermal patch (6 mg/24 hours) may be comparable to that of low-dose oral selegiline (10 mg) with neither causing a hypertensive reaction to high-tyramine meals
- Tyramine sensitivity and hypertensive responses to the high-dose transdermal patch (12 mg/24 hours) may occur with administration of 70–100 mg of tyramine in the fasted state, so dietary restrictions may

also not be necessary at 9 mg/24 hours or 12 mg/24 hours of transdermal administration of selegiline

- However, insufficient studies have been performed to be sure of the safety of transdermal administration at 9 mg/24 hours or 12 mg/24 hours
- Oral administration of nonselective irreversible MAOIs generally requires adherence to a low-tyramine diet (see [Table 2](#) after Pearls)
- Ingestion of a “high-tyramine meal,” defined as 40 mg or more of tyramine in the fasted state or as little as ingestion of 10 mg of tyramine in the fasted state, can cause hypertensive reactions in patients taking a nonselective irreversible MAOI orally
- Food restrictions (see [Table 2](#) after Pearls) are generally recommended for patients taking the higher doses of transdermal selegiline (9 mg/24 hours or 12 mg/24 hours) but not for the lower doses of transdermal selegiline (6 mg/24 hours) or for the low dose orally (10 mg)
- Transdermal: studies of low-dose transdermal administration of selegiline (6 mg/24 hours) failed to show changes in systolic or diastolic blood pressure or pulse when administered to normal volunteers taking either pseudoephedrine 60 mg three times a day for 2 days or 25 mg of phenylpropanolamine (no longer commercially available in the USA) every 4 hours for 1 day
- However, sufficient safety information is not available to recommend administration of pseudoephedrine without a precaution; blood pressure should be monitored if low-dose transdermal selegiline is given at all with pseudoephedrine

- Pseudoephedrine may need to be avoided when administering transdermal selegiline, particularly at higher doses of selegiline or in vulnerable patients with hypertension
- Although risk may be reduced with transdermal administration of selegiline, patient and prescriber must be vigilant to potential interactions with any drug, including antihypertensives and over-the-counter cough/cold preparations
- Over-the-counter medications to avoid or use with caution under the care of an expert include cough and cold preparations, including those containing dextromethorphan (a serotonin reuptake inhibitor), nasal decongestants (tablets, drops, or spray), hay-fever medications, sinus medications, asthma inhalant medications, anti-appetite medications, weight reducing preparations, “pep” pills (see [Table 3](#) after Pearls)
- Certain transdermal patches containing even small traces of aluminum or other metals in the adhesive backing can cause skin burns if worn during MRI, so warn patients taking the transdermal formulation about this possibility and advise them to disclose this information if they need an MRI
- Hypoglycemia may occur in diabetic patients receiving insulin or oral antidiabetic agents (more so with hydrazine MAOIs)
- Use cautiously in patients receiving reserpine, anesthetics, disulfiram, metrizamide, anticholinergic agents
- Selegiline is not recommended for use in patients who cannot be monitored closely

- Only use sympathomimetic agents or guanethidine with oral doses of selegiline below 10 mg/day

Table 2. Dietary guidelines for patients taking MAOIs*

Foods to avoid**	Foods allowed
Dried, aged, smoked, fermented, spoiled, or improperly stored meat, poultry, and fish	Fresh or processed meat, poultry, and fish; properly stored pickled or smoked fish
Broad bean pods	All other vegetables
Aged cheeses	Processed cheese slices, cottage cheese, ricotta cheese, yogurt, cream cheese
Tap and unpasteurized beer	Canned or bottled beer and alcohol
Marmite	Brewer's and baker's yeast
Banana peel	Bananas, avocados, raspberries
Sauerkraut, kimchee	Peanuts
Soy products/tofu	
Tyramine-containing nutritional supplement	

* Consult up-to-date listing of tyramine content, as most foods now have low tyramine amounts

** Not necessary for 6 mg transdermal or low-dose oral selegiline

Table 3. Drugs that boost norepinephrine: should only be used with caution with MAOIs

Use With Caution:			
Decongestants	Stimulants	Antidepressants with norepinephrine reuptake inhibition	Other
Phenylephrine	Amphetamines	Most tricyclics	Phentermine
Pseudoephedrine	Methylphenidate Cocaine Methamphetamine Modafinil Armodafinil	NRIs NDRI	Local anesthetics containing vasoconstrictors Tapentadol

Do Not Use

- If patient is taking meperidine (pethidine) (a serotonin reuptake inhibitor)
- If patient is taking a sympathomimetic agent or taking guanethidine
- If patient is taking another MAOI
- If patient is taking any agent that can inhibit serotonin reuptake (e.g., SSRIs, tramadol, milnacipran, duloxetine, venlafaxine, clomipramine, imipramine, etc.)
- If patient is taking dextromethorphan (a serotonin reuptake inhibitor)

- If patient is taking diuretics or other antihypertensives without careful monitoring of blood pressure
- If patient is taking St. John's wort, cyclobenzaprine, methadone, propoxyphene
- If patient has pheochromocytoma
- If patient is undergoing elective surgery and requires general anesthesia
- If there is a proven allergy to selegiline

Special Populations

Renal Impairment

- No dose adjustment necessary for transdermal administration in patients with mild to moderate renal impairment
- Use oral administration with caution –drug may accumulate in plasma in patients with renal impairment
- Oral administration may require lower than usual adult dose

Hepatic Impairment

- No dose adjustment necessary for transdermal administration in patients with mild to moderate hepatic impairment
- Oral administration may require lower than usual adult dose

Cardiac Impairment

- May require lower than usual adult dose
- Observe closely for orthostatic hypotension

Elderly

- Recommended dose for patients over 65 years old is 20 mg oral and 6 mg/day transdermal
- Dose increases in the elderly should be made with caution and patients should be observed for postural changes in blood pressure throughout treatment
- Reduction in the risk of suicidality with antidepressants compared to placebo in adults age 65 and older



Children and Adolescents

- Not recommended for use in children under 18
- Use with caution, observing for activation of known or unknown bipolar disorder and/or suicidal ideation, and inform parents or guardians of this risk so they can help observe child or adolescent patients
- Carefully weigh the risks and benefits of pharmacological treatment against the risks and benefits of nontreatment with antidepressants and make sure to document this in the patient's chart
- Monitor patients face-to-face regularly, particularly during the first several weeks of treatment



Pregnancy

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001
- Controlled studies have not been conducted in pregnant women
- Not generally recommended for use during pregnancy, especially during first trimester
- Should evaluate patient for treatment with an antidepressant with a better risk/benefit ratio
- National Pregnancy Registry for Antidepressants: 1-866-961-2388, <https://womensmentalhealth.org/research/pregnancyregistry/antidepressants>

Breast Feeding

- Some drug is found in mother's breast milk
- Immediate postpartum period is a high-risk time for depression, especially in women who have had prior depressive episodes, so drug may need to be reinstituted late in the third trimester or shortly after childbirth to prevent a recurrence during the postpartum period
- Should evaluate patient for treatment with an antidepressant with a better risk/benefit ratio

The Art of Psychopharmacology

Potential Advantages

- Treatment-resistant depression
- Patients with atypical depression (hypersomnia, hyperphagia)
- Patients who wish to avoid weight gain and sexual dysfunction
- Parkinson's patients inadequately responsive to levodopa

Potential Disadvantages

- Noncompliant patients
- Patients with motor complications and fluctuations on levodopa treatment

Primary Target Symptoms

- Depressed mood (depression)
- Somatic symptoms (depression)
- Sleep and eating disturbances (depression)
- Psychomotor disturbances (depression)
- Motor symptoms (Parkinson's disease)

Pearls

- Transdermal administration may allow freedom from dietary restrictions

- Transdermal selegiline theoretically appealing as a triple action agent (serotonin, norepinephrine, and dopamine) for treatment-refractory and difficult cases of depression
- Transdermal selegiline may have low risk of weight gain and sexual dysfunction, and may be useful for cognitive dysfunction in attention deficit disorder and other cognitive disorders, as it increases dopamine and is metabolized to l-amphetamine and l-methamphetamine
- Low-dose oral administration generally used as an adjunctive treatment for Parkinson's disease after other drugs have lost efficacy
- At oral doses used for Parkinson's disease, virtually no risk of interactions with food
- Neuroprotective effects are possible but unproved
- * Enhancement of levodopa action can occur for Parkinson's patients at low oral doses, but antidepressant actions probably require high oral doses that do not have the potential tissue selectivity and lack of dietary restrictions of the low-dose transdermal formulation
- * High doses may lose safety features
- MAOIs are generally reserved for third- or fourth-line use after SSRIs, SNRIs, and combinations of newer antidepressants have failed
- Patient should be advised not to take any prescription or over-the-counter drugs without consulting their doctor because of possible drug interactions
- Headache is often the first symptom of hypertensive crisis

- Myths about the danger of dietary tyramine can be exaggerated, but prohibitions against concomitant drugs often not followed closely enough
- * Combining multiple psychotropic agents with MAOIs should be for the expert, especially if combining with agents of potential risk (e.g., stimulants, trazodone, TCAs)
- * MAOIs should not be neglected as therapeutic agents for the treatment-resistant

Suggested Reading

Bodkin JA , Amsterdam JD . Transdermal selegiline in major depression: a double-blind, placebo-controlled, parallel-group study in outpatients . *Am J Psychiatry* 2002 ;**159** (11):1869–75 .

Gillman K . PsychoTropical Research (PTR). <https://psychotropic.com> .

Gillman PK . A reassessment of the safety profile of monoamine oxidase inhibitors: elucidating tired old tyramine myths . *J Neural Transm (Vienna)* 2018 ;**125** (11):1707–17 .

Kennedy SH . Continuation and maintenance treatments in major depression: the neglected role of monoamine oxidase inhibitors . *J Psychiatry Neurosci* 1997 ;**22** :127–31 .

Sertindole

Therapeutics

Brands

- Serdolect
- *see index for additional brand names*

Generic?

No

Class

- Neuroscience-based Nomenclature: dopamine and serotonin receptor antagonist (DS-RAn)
- Atypical antipsychotic (serotonin-dopamine antagonist; second-generation antipsychotics; also a mood stabilizer)

Commonly Prescribed for

(bold for FDA approved)

- Schizophrenia (for patients intolerant to at least one other antipsychotic)
- Acute mania/mixed mania
- Other psychotic disorders

- Bipolar maintenance
- Bipolar depression
- Treatment-resistant depression

How the Drug Works

- Blocks dopamine 2 receptors, reducing positive symptoms of psychosis and stabilizing affective symptoms
- Blocks serotonin 2A receptors, causing enhancement of dopamine release in certain brain regions and thus reducing motor side effects and possibly improving cognitive and affective symptoms
- * Serotonin 2C properties may contribute to antidepressant actions

How Long Until It Works

- Psychotic symptoms can improve within 1 week, but it may take several weeks for full effect on behavior as well as on cognition
- Classically recommended to wait at least 4–6 weeks to determine efficacy of drug, but in practice some patients may require up to 16–20 weeks to show a good response, especially on negative or cognitive symptoms

If It Works

- Most often reduces positive symptoms but does not eliminate them
- Can improve negative symptoms, as well as aggressive, cognitive, and affective symptoms in schizophrenia

- Most schizophrenia patients do not have a total remission of symptoms but rather a reduction of symptoms by about a third
- Perhaps 5–15% of schizophrenia patients can experience an overall improvement of greater than 50–60%, especially when receiving stable treatment for more than a year
- Such patients are considered super-responders or “awakeners” since they may be well enough to be employed, live independently, and sustain long-term relationships
- Continue treatment until reaching a plateau of improvement
- After reaching a satisfactory plateau, continue treatment for at least a year after first episode of psychosis
- For second and subsequent episodes of psychosis, treatment may need to be indefinite
- Even for first episodes of psychosis, it may be preferable to continue treatment

If It Doesn't Work

- Try one of the other atypical antipsychotics
- If 2 or more antipsychotic monotherapies do not work, consider clozapine
- Some patients may require treatment with a conventional antipsychotic